

Case Study 2:

Data Science at Golden State Warriors

JADE WINEBRIGHT

Problem Introduction



Ticket Timing Model

- Began 2019 off-season
- Ray Yocke, Senior Director of Business Analytics and Strategy
- ML model to predict when each fan could be swayed to purchase
- Originally promoted to mailing list through mass emails



Goals

- **Higher ticket sales**
- **Fewer emails**
- **Ideally – sell out arena before start of each game**



Data stored across multiple systems in different formats



Yocke's team focused only on direct ticket sales data

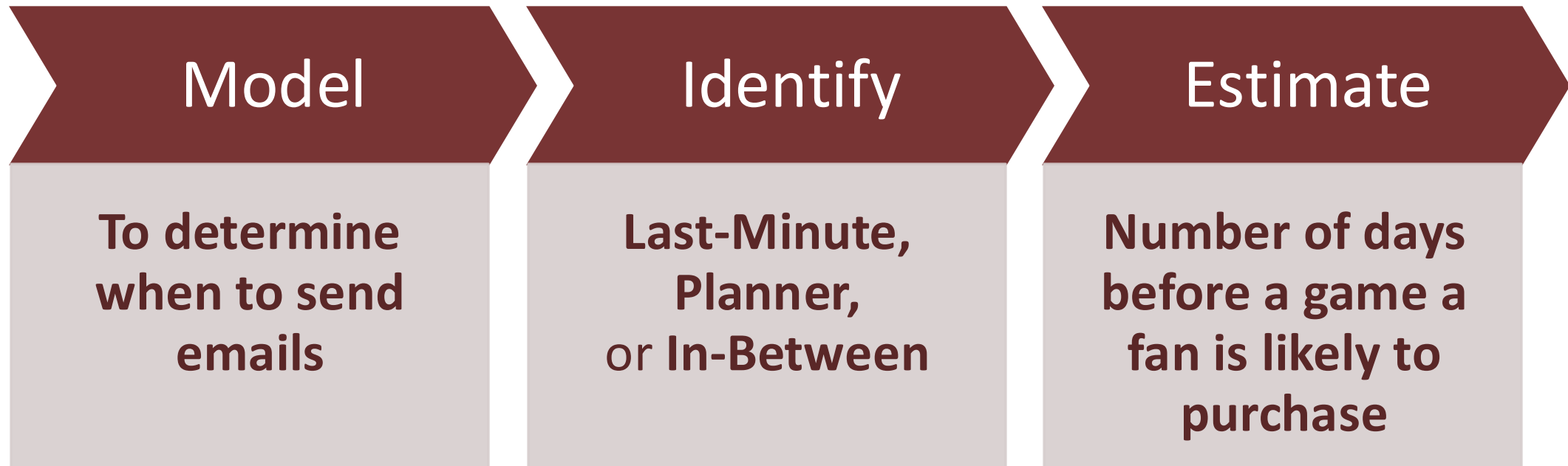


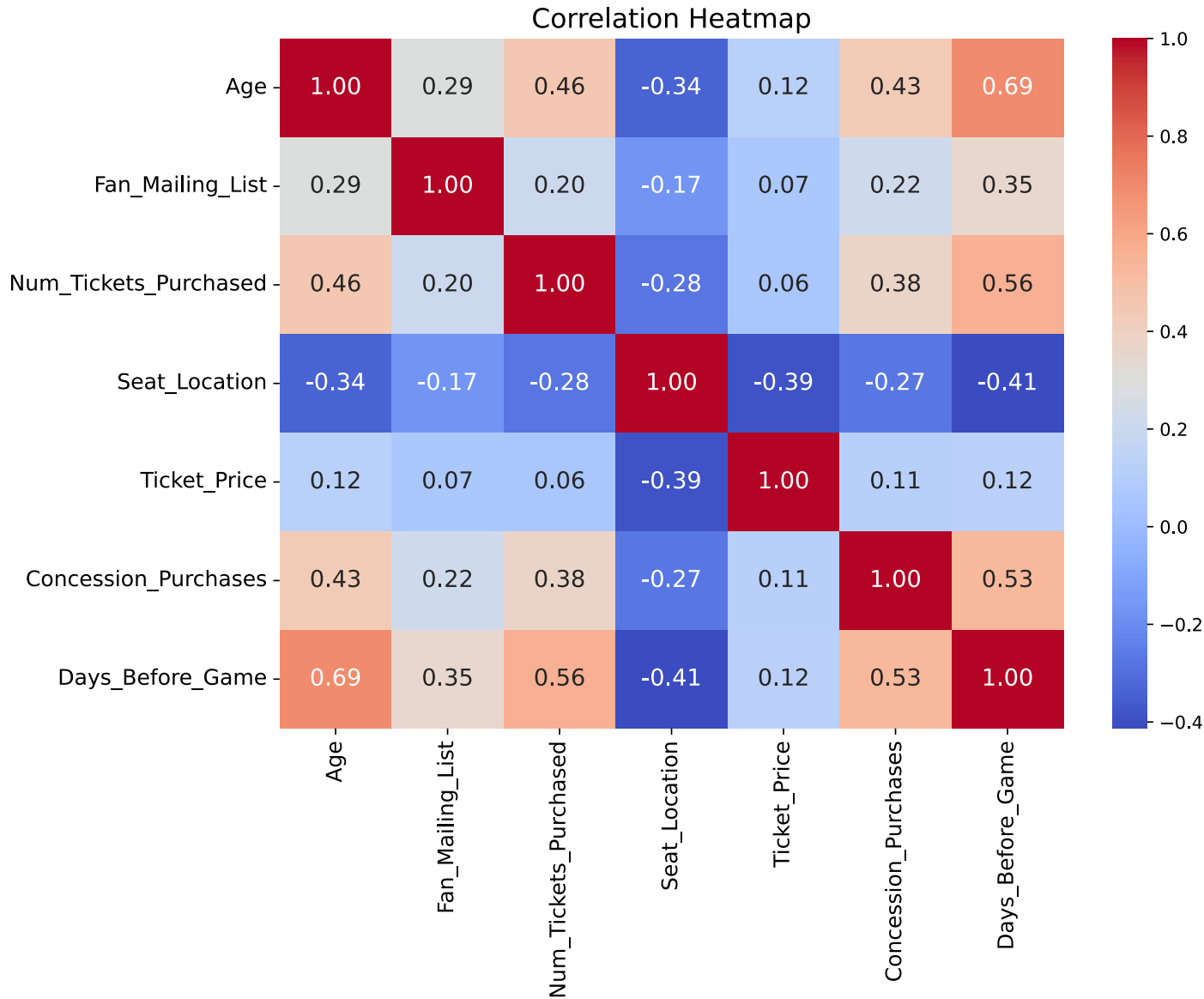
COVID related issues

Analysis of the Problem

- Possible Concerns

Approach





Approach

Correlations between features.

Looking at Days Before Game:

- Age (0.69)
- Tickets Purchased (0.56)
- Concession Purchases (0.53)
- Seat Location (-0.41)

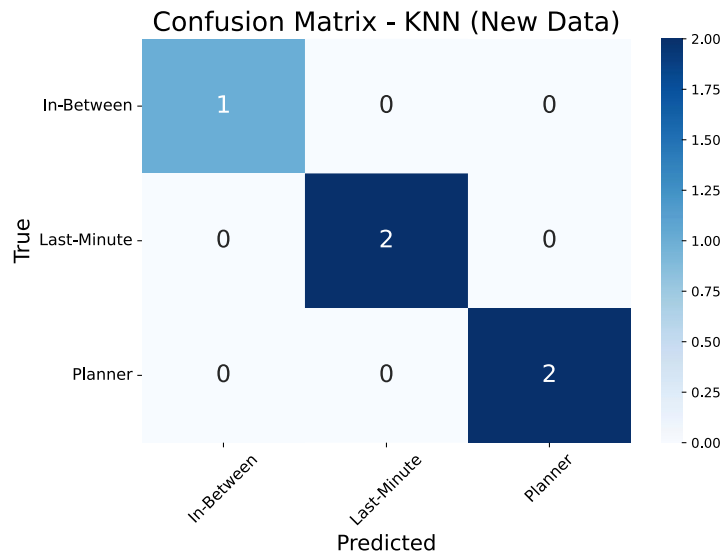
Goal 1: Classification

Goal: Can we improve the model that classifies purchasers into Last-Minute, Planners, or In-Between?

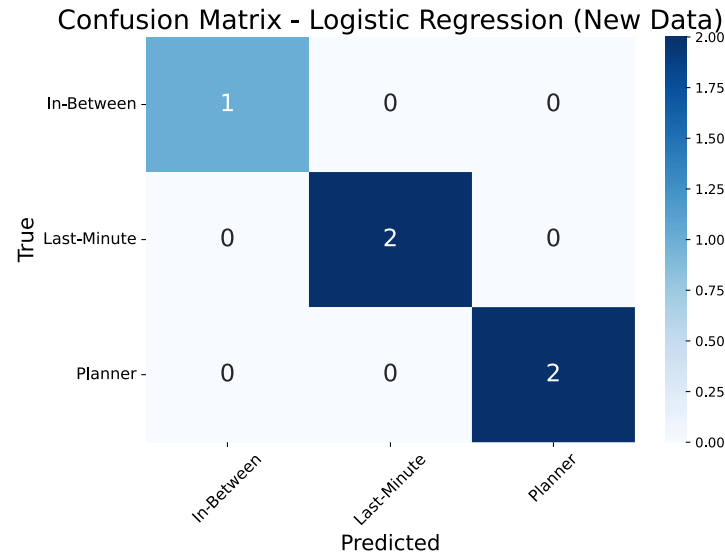
Tried 3 models:

- K-Nearest Neighbors (KNN)
- Logistic Regression
- Random Forest

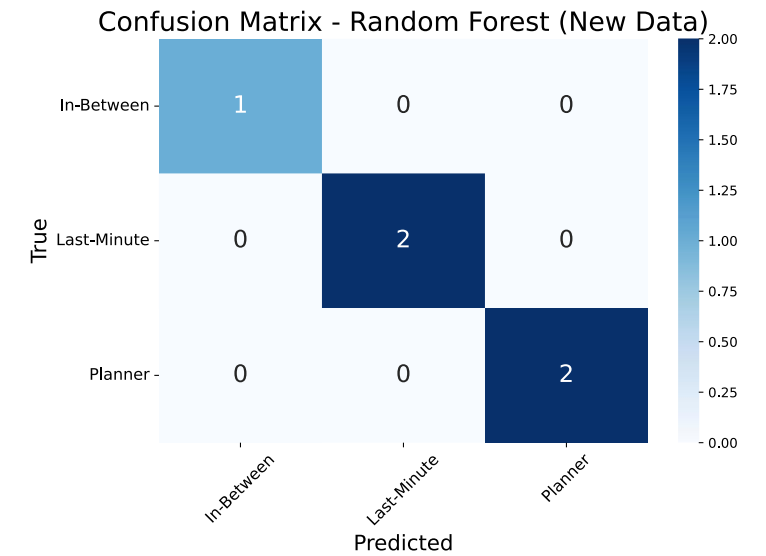
KNN



Logistic Regression



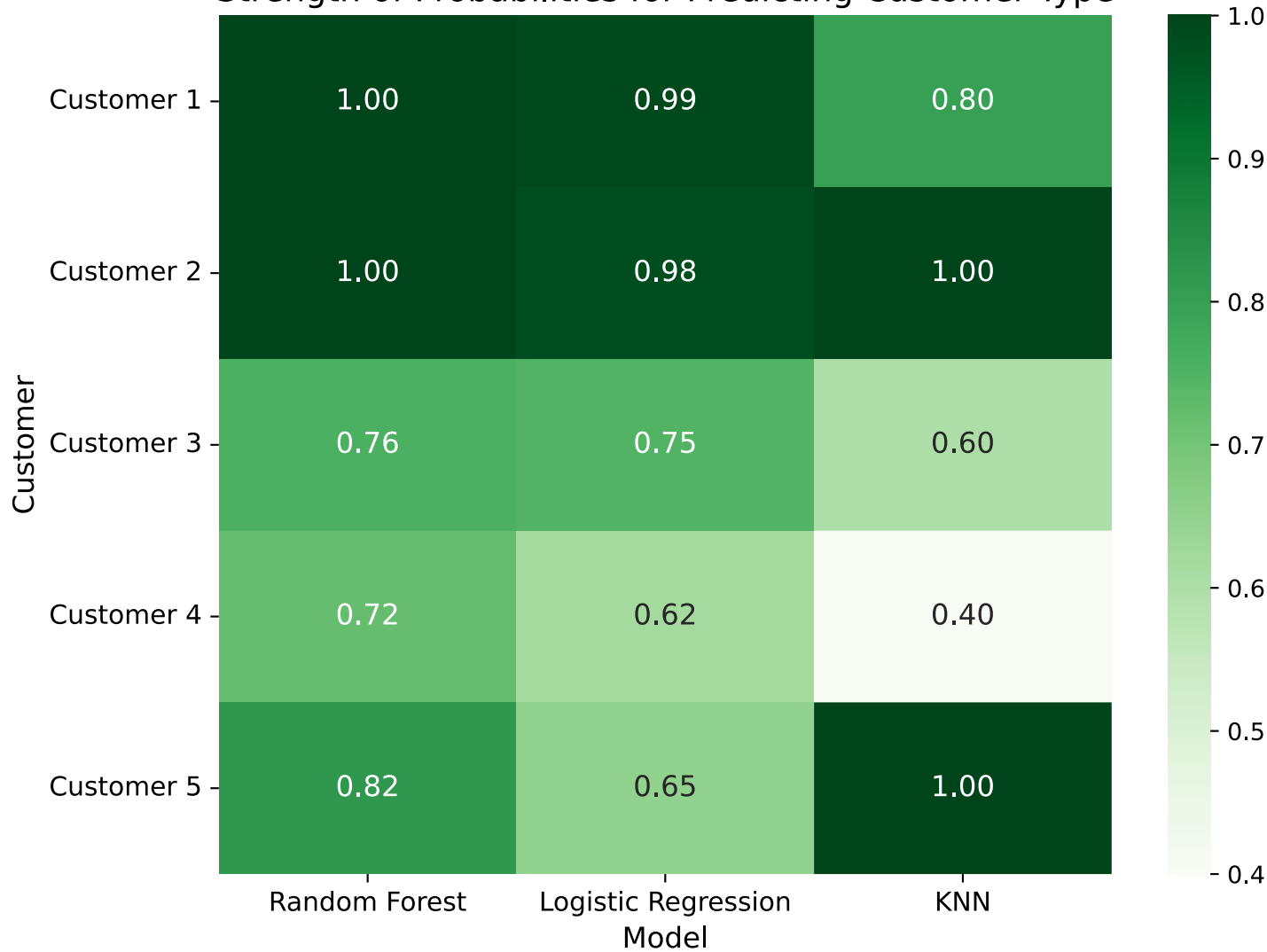
Random Forest



Findings

All models had 100% accuracy on the five test samples, so other metrics can be used to determine the best.

Strength of Probabilities for Predicting Customer Type



Model	Validation Accuracy
KNN	0.813
Logistic Regression	0.79
Random Forest	0.82

Random forest – highest validation accuracy of **0.82**

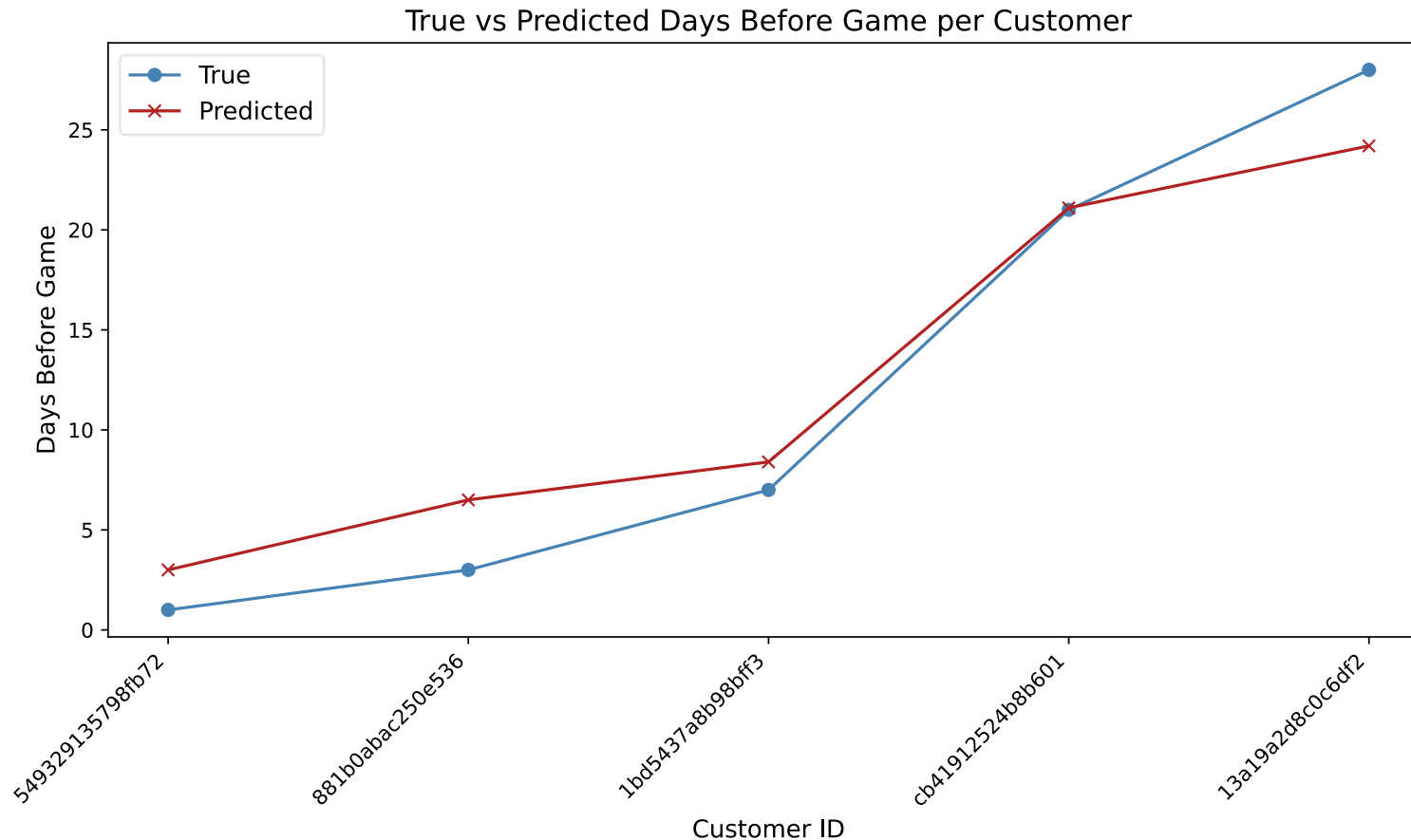
Random forest – highest probabilities/confidence in predictions (average of 0.86)

Goal 2: Regression

Goal: Can we predict the number of days that a customer is likely to purchase?

Random Forest Regression was used with parameter tuning.

Findings



True Days Before Game	Predicted Days Before Game
28	22.0
1	2.3
3	6.4
7	9.4
21	21.2

Using Random Forest Regressor Predictions for the five held-out test cases.

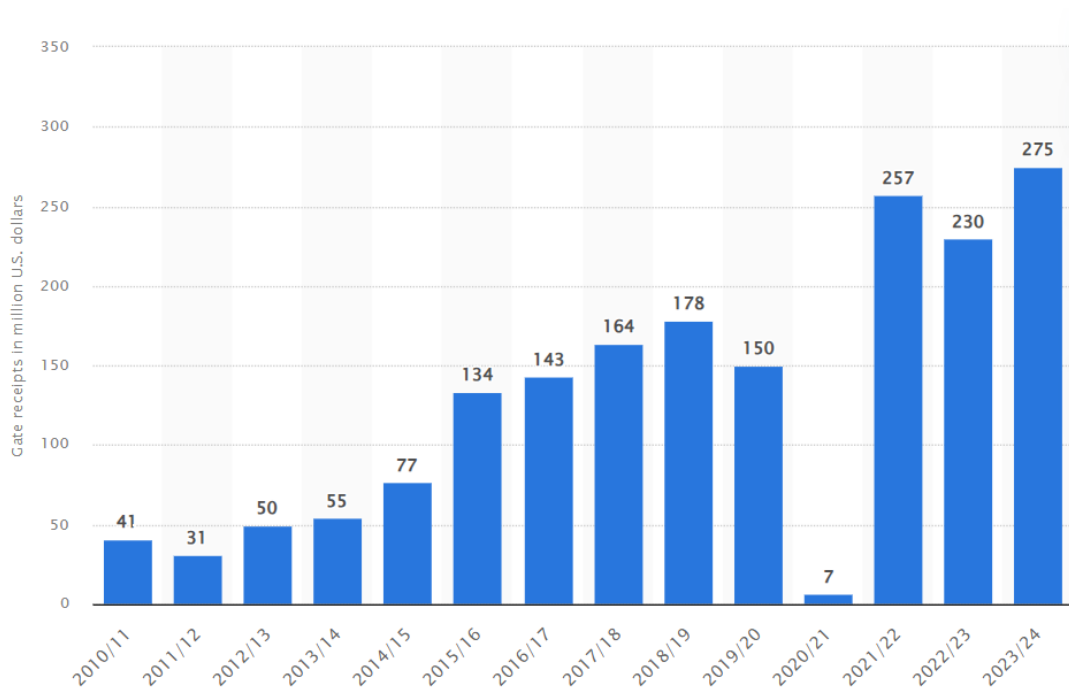
Findings

For predicting number of days, Random Forest (RF) regression performed better than the two previous models (Simple and Multiple Linear Regression).

RF's Mean Squared Error (MSE) for the 5 test samples was significantly lower.

Simple Linear Regression MSE	Multiple Linear Regression MSE	Best RF MSE
79.5456	20.0589	11.1764

Limitations



GSW gate receipts from 2010/11 to 2023/24 in millions

- In dataset, don't know much context (such as which game).
- Don't have specific financial information, only the total gate receipt sales for entire seasons.
- Both Oracle Arena & Chase Center hold about 18,000-19,000 fans. We were given 1,000 samples.
 - Context: Average number of tickets purchased was **2.8**, so we were only given about **15%** of arena capacity of a single game

Verification and Validation



Verification

- Preprocessing steps verified through EDA and visual inspection
- Models constructed using standard libraries
 - Code outputs manually reviewed
 - Inspected to ensure no data leakage



Validation

- Designed to improve the current Ticket Timing Model
- Models followed characteristics of real-world data
 - Results statistically sound and practically useful
 - Interpretable and comparable results



Feature Relationships – age, number of tickets, seat location, and concession purchases were most correlated with purchase timing



Fan-Type Classification – Random Forest performed better than Logistic Regression, KNN, or the current models



Specific Purchase Timing Regression – Random Forest with parameter tuning performed better than current models

Conclusions

Recommendations

Modeling

Use Random Forest Classification and Random Forest Regression for predicting both type of fan and exact number of days, use this information for future email timing.

Expand data collection

More games, more customers, specific game details, pre-purchase indicators such as website visits.

Targeted Campaigns

Roughly 29% of buyers purchase within two days of game day. Use time-sensitive campaigns (inventory alerts, countdown messaging, pricing incentives).